# Run a Simulation

Set your parameters and simulate 10,000 possible market futures.

Simple | Risk Analysis

Risk Analysis mode adds jump shocks, market regime switching, and volatility clustering.
Plain GBM does not model tail risk — this mode does.

PRESET SCENARIO

Early Retirement

Initial Investment $25,000

$0 — $100,000

Monthly Contribution $1,500

$0 — $5,000

Time Horizon 25 years

5 yrs — 40 yrs

Return Profile

Conservative 5% avg | Moderate 7% avg | Aggressive 10% avg

Volatility 22.0%

2% — 60%

Adjust for Inflation
Shows real purchasing power

Jump Events

Crash Frequency (per year) 0.5x

Avg Crash Size −25.0%

Crash Volatility 10.0%

Market Regimes

Volatility Clustering

After a crash, volatility spikes 2× and decays back to baseline over subsequent months. Captures the "vol begets vol" effect seen in real markets.

Set a goal

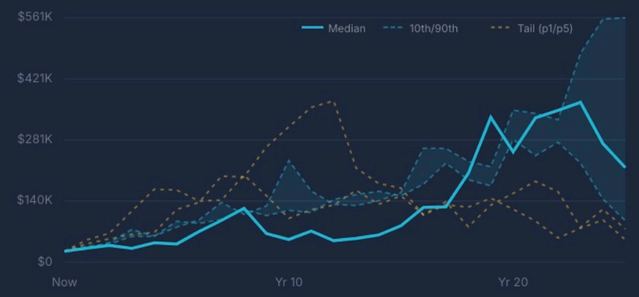
Target portfolio value — we'll show the probability of reaching it across all 10,000 paths

| Median Outcome | Best Case | Worst Case | Chance of Reaching Goal |
|---|---|---|---|
| $217K | $561K | $95K | 12.8% |
| 50th percentile | 90th percentile | 10th percentile | $500,000 |

| Avg Max Drawdown | Extreme Downside (p1) | Severe Downside (p5) |
|---|---|---|
| 69.4% | $51K | $76K |
| peak-to-trough avg | 1st percentile path | 5th percentile path |

## Portfolio Growth Paths

Representative simulation paths nearest the 10th, 50th, and 90th percentile terminal values. Amber lines show the 1st and 5th percentile (tail risk) paths.




## Distribution of Outcomes

Final portfolio values across all 10,000 simulations. Bright bars show simulations that reached your goal.

GBM + jump-diffusion + regime switching · 10,000 paths · monthly stepping · 3% inflation adjustment